

Credal degrees of freedom for PLN truth values: a layered characterization of strength and confidence

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Abstract

PLN’s two-number truth value $\langle s, c \rangle$ is the compression of a richer object: a credal set of admissible precise previsions over an outcome space. We give a layered formal characterization of where $\langle s, c \rangle$ is *determined* by the spec — the strength is a single real, the confidence is at its information ceiling — and where the underlying credal set has *degrees of freedom* that the two-number summary cannot see.

The characterization rests on a recently-landed Lean dichotomy in `Mettapedia.ProbabilityTheory.ImpreciseProbability.ProjectiveCredal`: the predicates `credalSetDetermines` and `credalSetHasStrictWidth` on a credal set C and a gamble X are the formal locators of, respectively, “ $\langle s, c \rangle$ is fully determined on X ” and “ $\langle s, c \rangle$ has genuine interval width on X .” The collapse theorem `lower_eq_upperEnvelope_of_credalSetDetermines` closes the “determined $\Rightarrow$ precise” direction; the width theorem `lower_upperEnvelope_nontrivial_of_strictWidth` closes the “disagreement $\Rightarrow$ envelope opens” direction.

We lift this dichotomy to the projective credal level (`determinesGlobalGamble` / `hasStrictGlobalWidth`) and onward to the infinite Markov-logic setting through DLR completions (`dlrQueryDetermined` / `dlrQueryHasStrictWidth`). Under the Dobrushin uniform small-influence hypothesis, `PaperUniformSmallTotalInfluence` implies `dlrQueryDetermined`, and therefore the MLN query envelope is precise. In the opposite direction, whenever two DLR completions are explicitly exhibited that disagree on a finite query, the corresponding Walley envelope has strict width. Failure of the Dobrushin hypothesis is therefore a frontier where strict width can occur, not by itself a proof that it must occur.

We organize the degrees of freedom into three slots: (A) specification-level freedom in the local credal sets; (B) completion-level freedom in extreme-point choice within a fixed credal set (aleatoric, formally invisible to $\langle s, c \rangle$); (C) phase-transition freedom from non-uniqueness of DLR completions in infinite MLNs. We honestly enumerate what is open.

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1 Introduction

PLN’s atomic truth-value object is a pair $\langle s, c \rangle$ with $s, c \in [0, 1]$ called *strength* and *confidence*. Standard PLN parametrises s as a probability estimate and c as a function of evidence count, typically $c = N/(N + \kappa)$ for a system-level constant κ and an evidence-count integer N . The pair is what travels through PLN’s rules (revision, deduction, induction, abduction); it is what is read off by query atoms; it is the type signature of every PLN belief.

But $\langle s, c \rangle$ is a two-number summary. The same pair can be produced by very different underlying epistemic objects: a Beta-distribution second-order Bayesian; a Walley-style coherent lower prevision with a specific interval width; an exchangeable de Finetti mixture with a particular concentration. PLN’s compositional rules treat $\langle s, c \rangle$ as the belief object, and the algebra reduces to numerical update formulas; but the question of *what is and is not pinned down by* $\langle s, c \rangle$ is rarely written explicitly in the PLN literature.

This paper makes that explicit. We organise the question as follows. The strength and confidence of a query in PLN are determined by an underlying credal set C of admissible precise previsions. The *degrees of freedom* that $\langle s, c \rangle$ summarises but does not expose live in three slots:

- Specification freedom in the local credal sets that build C ;
- Completion freedom (extreme-point choice) inside a fixed C ;
- Phase-transition freedom from non-uniqueness of global completions when C is the projective limit of a Markov-logic / DLR family.

Each slot has a formally locatable status. Recent landings in the `Mettapedia.ProbabilityTheory.ImpreciseProbability` and `Mettapedia.Logic` libraries (the *credal dichotomy* predicates and the DLR query lifts; see Sec. 4 and 6) make this dichotomy an actual Lean theorem rather than informal prose. Wherever possible we quote the Lean theorem statements verbatim from their source files.

Scope. We treat strength and confidence as the read-off of credal-envelope width and centre. We do *not* address the propagation of $\langle s, c \rangle$ through PLN’s compositional rules (deduction, induction, abduction, revision); that material is in the existing PLN corpus (ξ PLN, ν PLN, the WM-PLN book; see Sec. 10). The paper’s contribution is the determinacy / freedom characterization at the level of a single truth value, with formal Lean hooks.

Sibling papers. This paper is a sibling of ξ PLN and the *PLN Review* (`papers/xiPLN.tex`, `papers/pln-review.tex`). The WM-PLN book `papers/wm-pln-book.tex` contains an informal section on degrees of freedom versus forcing (§7.4 in the current draft); the present paper offers the formal characterization that the book section calls for.

2 The credal stack: types and definitions

We work in the formal stack of `Mettapededia.ProbabilityTheory.ImpreciseProbability.ProjectiveCredal` (referenced throughout as *ProjectiveCredal.lean*, file path `Mettapededia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean`). Outcome spaces are arbitrary types Ω ; gambles are bounded real-valued functions on Ω .

2.1 Gambles and previsions

A *gamble* on Ω is a real-valued function on Ω :

```
abbrev Gamble (Omega : Type*) := Omega -> Real
```

A *precise prevision* on Ω is a Walley-coherent linear functional from gambles to reals: the structure `PrecisePrevision` in `ProjectiveCredal.lean` is the standard linear expectation with constant-preservation, positive homogeneity, and additivity built in:

```
structure PrecisePrevision (Omega : Type*) where
  toFun      : Gamble Omega -> Real
  lower_bound : ... -- c <= X (pointwise) implies c <= toFun X
  pos_homog   : ... -- toFun (r . X) = r * toFun X for r >= 0
  add         : ... -- toFun (X + Y) = toFun X + toFun Y
```

A *lower prevision* relaxes additivity to superadditivity, and is the Walley credal-summary object:

```
structure LowerPrevision (Omega : Type*) where
  toFun      : Gamble Omega -> Real
  lower_bound : ... -- c <= X pointwise implies c <= toFun X
  pos_homog   : ... -- toFun (r . X) = r * toFun X for r >= 0
  superadd    : ... -- toFun (X + Y) >= toFun X + toFun Y
```

Coherence (positive homogeneity, superadditivity, and avoidance of sure loss) is the Walley condition for the lower prevision to be the lower envelope of some non-empty set of precise previsions.

2.2 Credal sets and envelopes

A *credal set* is just a set of precise previsions:

```
abbrev CredalPrevisionSet (Omega : Type*) := Set (PrecisePrevision
  Omega)
```

For a credal set C and a gamble X , the *lower* and *upper* envelopes are the standard inf / sup:

```
def lowerEnvelope (C : CredalPrevisionSet Omega) (X : Gamble Omega) :
  Real :=
  sInf ((fun P : PrecisePrevision Omega => P X) ' ' C)
```

```
def upperEnvelope (C : CredalPrevisionSet Omega) (X : Gamble Omega) :
  Real :=
  sSup ((fun P : PrecisePrevision Omega => P X) '' C)
```

The construction `lowerEnvelopePrevision` packages `lowerEnvelope` together with the proofs of constant-preservation, positive homogeneity, and superadditivity, exhibiting a `LowerPrevision` object out of any non-empty bounded-below credal set. The finite case theorem `lowerEnvelopePrevision_isCoherent_of_finite` (ProjectiveCredal.lean line 868) confirms Walley coherence on finite nonempty Ω .

2.3 PLN $\langle s, c \rangle$ as the credal-envelope summary

The canonical Lean STV type lives in `Mettapedia.Logic.PLNDeduction` at lines 138–139:

```
-- Simple Truth Value: (strength, confidence) pair -/
structure STV where
  strength      : Real
  confidence    : Real
  strength_nonneg : 0 <= strength
  strength_le_one : strength <= 1
  confidence_nonneg : 0 <= confidence
  confidence_le_one : confidence <= 1
```

Its distributional model is in `Mettapedia.Logic.PLNDistributional`: given a $\text{Beta}(\alpha, \beta)$ distribution, the canonical PLN STV reads

$$s = \frac{\alpha}{\alpha + \beta}, \quad c = f(\alpha + \beta)$$

for an evidence-count map f (typically $f(N) = N/(N + \kappa)$).

The two-number compression from a credal set C and a query gamble X_q (e.g. the indicator $\mathbf{1}_A$ of a query atom A) is the following implicit map:

$$C, X_q \mapsto \begin{cases} s_q = \frac{1}{2}(\text{lowerEnvelope } C X_q + \text{upperEnvelope } C X_q) \\ c_q = g(\text{upperEnvelope } C X_q - \text{lowerEnvelope } C X_q) \end{cases} \quad (1)$$

for a monotone-decreasing $g : [0, 1] \rightarrow [0, 1]$ with $g(0) = 1$ (no width, full confidence) and $g(1) = 0$ (maximum width, no information). The specific choice of midpoint and g is an interface design decision (PLN traditionally uses midpoint for s and a sigmoidal g derived from the Beta variance); the freedom characterization of this paper is independent of this choice. What matters is that (1) is the read-off, and that the credal set C may have degrees of freedom that the pair (s_q, c_q) cannot expose.

Remark 1. *The summary map (1) loses two kinds of information. First, which gambles have positive credal-width and which do not: (s_q, c_q) summarises a single query but a credal set with non-zero width on some queries may have zero width on others. Second, the shape of C : a credal set on Ω whose extreme points*

are clustered (high agreement) and one whose extreme points are dispersed (low agreement) can produce the same midpoint and width on a particular query, but compose differently under PLN's inference rules. The width as a function on gambles — which we will study below — is the right object; (s_q, c_q) is its evaluation at one point.

3 Finite-case determinacy

We first treat the case where the outcome space Ω is finite. Here the Walley theory is complete in the formalization; we summarise the load-bearing results.

3.1 Uniform positivity and avoidance of sure loss

For finite Ω , a pointwise-strictly-positive gamble has a uniform positive lower bound (ProjectiveCredal.lean line 833):

```
theorem finite_strictlyPositive_uniformLowerBound
  {Omega : Type*} [Fintype Omega] [Nonempty Omega]
  (X : Gamble Omega) (hX : forall omega, 0 < X omega) :
  exists epsilon : Real, 0 < epsilon and forall omega, epsilon <= X
    omega
```

This is the technical hinge that lets every nonempty finite credal set *avoid sure loss* in the strict Walley sense (line 855):

```
theorem lowerEnvelopePrevision_avoidsSureLoss_of_finite
  {Omega : Type*} [Fintype Omega] [Nonempty Omega]
  (C : CredalPrevisionSet Omega) (hC : C.Nonempty)
  (hBdd : forall X : Gamble Omega,
    BddBelow ((fun P : PrecisePrevision Omega => P X) '' C)) :
  (lowerEnvelopePrevision C hC hBdd).avoidsSureLoss
```

and to be *coherent* (line 868):

```
theorem lowerEnvelopePrevision_isCoherent_of_finite
  {Omega : Type*} [Fintype Omega] [Nonempty Omega]
  (C : CredalPrevisionSet Omega) (hC : C.Nonempty)
  (hBdd : forall X : Gamble Omega,
    BddBelow ((fun P : PrecisePrevision Omega => P X) '' C)) :
  (lowerEnvelopePrevision C hC hBdd).isCoherent
```

The take-away for PLN: on a finite outcome space, the credal-envelope construction never fails. The map from spec to $\langle s, c \rangle$ is total.

3.2 Singleton collapse

The case where the credal set has exactly one element is the case where PLN's $\langle s, c \rangle$ has no freedom at all; everything is pinned. The theorem (ProjectiveCredal.lean line 885) is direct:

```

theorem lowerEnvelope_singleton
  {Omega : Type*} (P : PrecisePrevision Omega) (X : Gamble Omega) :
  lowerEnvelope ({P} : CredalPrevisionSet Omega) X = P X

```

and similarly for `upperEnvelope_singleton`. On a singleton credal set, lower = upper on every gamble. PLN-side: every query has c at its information ceiling, and s is the single value PX_q .

4 The credal dichotomy: where strength is determined and where it has width

The contribution of the recent landing in `ProjectiveCredal.lean` is to make the “determined / has width” distinction *predicate-level*, not theorem-level. Two definitions are added (lines 746 and 753), each followed by collapse / non-triviality theorems that pin down the PLN-side consequence.

4.1 The two predicates

Definition 2 (determined credal set on a gamble). *The predicate `credalSetDetermines` (`ProjectiveCredal.lean` line 746):*

```

def credalSetDetermines {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega) : Prop :=
  forall P : PrecisePrevision Omega, P in C ->
  forall Q : PrecisePrevision Omega, Q in C -> P X = Q X

```

C determines X when every two admissible precise previsions in C assign the same expectation to X .

Definition 3 (strict width). *The predicate `credalSetHasStrictWidth` (line 753):*

```

def credalSetHasStrictWidth {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega) : Prop :=
  exists P : PrecisePrevision Omega, P in C and
  exists Q : PrecisePrevision Omega, Q in C and P X < Q X

```

C has strict width on X when there exist two precise previsions in C that strictly disagree on X .

The two predicates are mutually exclusive (`ProjectiveCredal.lean` line 809):

Theorem 4 (mutual exclusion). *`not_credalSetDetermines_of_strictWidth`:*

```

theorem not_credalSetDetermines_of_strictWidth
  {Omega : Type*}
  {C : CredalPrevisionSet Omega} {X : Gamble Omega}
  (hWidth : credalSetHasStrictWidth C X) :
  not (credalSetDetermines C X)

```

The endpoint instances cover the trivial cases. Subsingleton credal sets collapse to determinacy (`credalSetDetermines_of_subsingleton`, line 758), and so do singletons (`credalSetDetermines_singleton`, line 766):

```
theorem credalSetDetermines_singleton {Omega : Type*}
  (P : PrecisePrevision Omega) (X : Gamble Omega) :
  credalSetDetermines ({P} : CredalPrevisionSet Omega) X
```

4.2 Collapse and width theorems

The PLN-side consequence of `credalSetDetermines` is precisely that strength is pinned: the lower and upper envelopes coincide (`ProjectiveCredal.lean` line 798):

Theorem 5 (envelope collapse from determinacy). *lower_eq_upperEnvelope_of_credalSetDetermines:*

```
theorem lower_eq_upperEnvelope_of_credalSetDetermines
  {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega)
  (hC : C.Nonempty)
  (hBddBelow : BddBelow ((fun P => P X) '' C))
  (hBddAbove : BddAbove ((fun P => P X) '' C))
  {P : PrecisePrevision Omega} (hP : P in C)
  (hDet : credalSetDetermines C X) :
  lowerEnvelope C X = upperEnvelope C X
```

This is the formal statement of “ $\langle s, c \rangle$ on X_q is precise” when C determines X_q .

The PLN-side consequence of `credalSetHasStrictWidth` is the opposite: the envelope is genuinely non-trivial (line 835):

Theorem 6 (envelope width from strict disagreement). *lower_upperEnvelope_nontrivial_of_strictWidth:*

```
theorem lower_upperEnvelope_nontrivial_of_strictWidth
  {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega)
  (hBddBelow : BddBelow ((fun P => P X) '' C))
  (hBddAbove : BddAbove ((fun P => P X) '' C))
  (hWidth : credalSetHasStrictWidth C X) :
  lowerEnvelope C X < upperEnvelope C X
```

This is the formal statement of “ $\langle s, c \rangle$ on X_q is genuinely an interval” when two completions in C strictly disagree on X_q .

4.3 PLN read-off

Putting the dichotomy into the summary map (1): for any credal set C on which both envelope-boundedness conditions hold, and any query gamble X_q , exactly one of the following holds:

predicate	envelope	PLN $\langle s, c \rangle$
<code>credalSetDetermines $C X_q$</code>	$L = U$	precise $s = L X_q$; c at ceiling
<code>credalSetHasStrictWidth $C X_q$</code>	$L < U$	s in interval $[L, U]$; $c < \text{ceiling}$

The Lean dichotomy is the formal locator of which case the PLN truth-value read-off lives in.

5 Projective credal specs: three slots for freedom

Once Ω is large (or infinite), one builds the credal set *compositionally* from finite-window data. The standard packaging is a projective family of local credal sets, indexed by “windows” (finite subsets, finite cylinders) of the global outcome space, glued by restriction maps.

5.1 The projective machinery

The cylinder structure (ProjectiveCredal.lean, the ProjectiveCylinderSystem structure) packs the window-to-local projection and the local restriction maps:

```
structure ProjectiveCylinderSystem (Window Global : Type*) [LE Window
] where
  Local          : Window -> Type*
  project        : forall i : Window, Global -> Local i
  restrict       : forall {i j : Window}, i <= j -> Local j ->
    Local i
  project_restrict : ... -- compatibility on the diagram
```

The full spec layers a local credal choice on top:

```
structure ProjectiveLocalCredalSpec (Window Global : Type*) [LE
Window] where
  cylinders      : ProjectiveCylinderSystem Window Global
  localCredal    : forall i : Window, CredalPrevisionSet (cylinders.
    Local i)
```

The *global credal set* associated to a spec is the projectiveLimitCredalSet (the set of P in PrecisePrevision Global that marginalise into localCredal i at every window i).

5.2 Three identifying cases and the dichotomy at the global level

The codebase ships two endpoints and one general construction:

- **Singleton** (ProjectiveCredal.lean line 1615):

```
def singletonIdentityProjectiveSpec {Omega : Type*}
  (P : PrecisePrevision Omega) : ProjectiveLocalCredalSpec
  PUnit Omega
```

Every window’s local credal is the supplied singleton $\{P\}$. The projective limit is $\{P\}$ (`singletonIdentityProjectiveSpec_projectiveLimitCredalSet`, line 1620). PLN $\langle s, c \rangle$ on every gamble is precise; this is the “no freedom” case.

- **Unrestricted** (line 1571):

```
def finiteUnrestrictedProjectiveSpec (Omega : Type*) :
  ProjectiveLocalCredalSpec PUnit Omega
```

Every window’s local credal is `Set.univ`. The projective limit contains every precise prevision on Ω . PLN $\langle s, c \rangle$ on most gambles has `credalSetHasStrictWidth`; this is the “maximum freedom” case. The non-emptiness of the projective limit is witnessed by `finiteUnrestricted_finiteEvaluationCompact_hasCompatibleCompletion` (line 1578) via a compact carrier in the finite-evaluation topology.

- **Genuine intermediate:** any spec whose local credals are non-trivial proper subsets of `Set.univ`. Whether `credalSetDetermines` or `credalSetHasStrictWidth` holds on a particular gamble depends on the spec and the gamble.

The dichotomy lifts to the projective level. `ProjectiveCredal.lean` defines `determinesGlobalGamble` (line 1307) and `hasStrictGlobalWidth` (line 1314) as the global analogues of `credalSetDetermines` and `credalSetHasStrictWidth`, with the mutual exclusion theorem `not_determinesGlobalGamble_of_strictWidth` at line 1411. The natural-extension lifting theorems (around lines 1391 and 1424) connect the projective predicates to the lower / upper envelopes on the projective limit.

5.3 The three formal slots, structurally

We can now name the three slots in which PLN credal freedom lives. Each is formally locatable through one of the predicates above.

- (A) **Specification-level freedom.** The choice of `localCredal` i at each window. This is the modeller’s freedom: evidence width, prior choices, and what classes of measure are taken to be admissible at each local scale.
- (B) **Completion-level freedom.** Given a fixed credal set C , the choice of $P \in C$ that actually governs the world. This freedom is purely aleatoric: it is invisible to $\langle s, c \rangle$ because $\langle s, c \rangle$ is a property of C , not of any specific $P \in C$.

(C) **Phase-transition freedom.** For globally-glued projective specs, the projective limit may itself have multiple admissible elements that correspond to different *global* solutions of the same local problem. This is the slot Markov-logic / DLR exemplifies (next section).

Slots (A) and (C) are the only ones $\langle s, c \rangle$ can see. Slot (B) is formally invisible at the truth-value level.

6 Infinite Markov logic: Dobrushin collapse and DLR width

For the infinite case — countably-many ground atoms, Markov-logic network with weights, density specified locally — the projective machinery specialises into the Dobrushin–Lanford–Ruelle framework. The file `Mettapedia.Logic.MarkovLogicInfiniteCredalBridge` (Mettapedia/Logic/MarkovLogicInfiniteCredalBridge.lean) packages this.

6.1 Classical MLN spec and DLR completions

A `ClassicalInfiniteGroundMLNSpec` packages a countable atom set, a clause-id set, the clauses, their log-weights, and the local finite-volume region structure. A *DLR completion* of such a spec is a probability measure on the infinite world satisfying the DLR consistency condition at every finite volume:

```
abbrev DLRCompletion (M : ClassicalInfiniteGroundMLNSpec Atom
  ClauseId) :=
  { mu : ProbabilityMeasure (InfiniteWorld Atom)
    // FixedRegionCylinderDLR M.toStrictlyPositiveInfiniteGroundMLNSpec
    (mu : Measure (InfiniteWorld Atom)) }
```

The set of DLR completions of M is the “credal set” for the infinite spec, and the projective-credal lift packages it as `DLRProjectiveCredalSpecialization` in `Mettapedia/Logic/MarkovLogicInfiniteCredalBridge.lean`.

A *query* is a finite-volume event; its *probability* under a DLR completion is its measure:

```
noncomputable def dlrCompletionQueryProb
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom) (mu : DLRCompletion M) : Real :=
  ENNReal.toReal
  ((infiniteMLNMassSemantics M mu.1 mu.2).queryProb q)
```

6.2 DLR-level dichotomy

The credal dichotomy lifts directly to the DLR setting:

```
def dlrQueryDetermined
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
```

```

(q : ConstraintQuery Atom) : Prop :=
  forall mu nu : DLRCompletion M,
    dlrCompletionQueryProb M q mu = dlrCompletionQueryProb M q nu

def dlrQueryHasStrictWidth
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom) : Prop :=
  exists mu : DLRCompletion M, exists nu : DLRCompletion M,
    dlrCompletionQueryProb M q mu < dlrCompletionQueryProb M q nu

```

`dlrQueryDetermined` says all DLR completions agree on the query; `dlrQueryHasStrictWidth` says two of them strictly disagree. The mutual exclusion theorem `not_dlrQueryDetermined_of_strictWidth`:

```

theorem not_dlrQueryDetermined_of_strictWidth
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom)
  (hWidth : dlrQueryHasStrictWidth M q) :
  not (dlrQueryDetermined M q)

```

6.3 Dobrushin uniqueness as the collapse-direction hypothesis

The classical Dobrushin uniform-influence condition is packaged as the predicate `PaperUniformSmallTotalInfluence` on the MLN spec in `Mettapedia.Logic.MarkovLogicInfiniteUniqueness`. The collapse direction of the DLR-level dichotomy is the theorem `dlrQueryDetermined_of_uniform`:

```

theorem dlrQueryDetermined_of_uniform
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (hM : M.PaperUniformSmallTotalInfluence)
  (q : ConstraintQuery Atom) :
  dlrQueryDetermined M q

```

Under the Dobrushin uniform small-influence hypothesis, every finite query is DLR-determined. Combining with the dichotomy gives the precision of the MLN query envelope:

```

theorem infiniteMLN_queryEnvelope_precise_of_uniform
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  [Nonempty (DLRCompletion M)]
  (hM : M.PaperUniformSmallTotalInfluence)
  (q : ConstraintQuery Atom) :
  infiniteMLNLowerQueryEnvelope M q
  = infiniteMLNUpperQueryEnvelope M q

```

Under the same hypothesis, and assuming the DLR completion family is inhabited, the PLN $\langle s, c \rangle$ for every finite query is precise.

6.4 Strict width as the phase-transition canary

The opposite direction is the theorem `infiniteMLN_queryEnvelope_nontrivial_of_disagreement` and the lift through `infiniteMLN_queryEnvelope_`

nontrivial_of_strictWidth:

```
theorem infiniteMLN_queryEnvelope_nontrivial_of_strictWidth
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom)
  (hBddBelow : BddBelow (Set.range (dlrCompletionQueryProb M q)))
  (hBddAbove : BddAbove (Set.range (dlrCompletionQueryProb M q)))
  (hWidth : dlrQueryHasStrictWidth M q) :
  infiniteMLNLowerQueryEnvelope M q
    < infiniteMLNUpperQueryEnvelope M q
```

If two DLR completions disagree on a finite query, the PLN $\langle s, c \rangle$ for that query is genuinely an interval.

6.5 Dobrushin collapse and strict-width witnesses

Combining the two proved directions gives a one-sided collapse theorem and a separate strict-width witness theorem. The Dobrushin condition `PaperUniformSmallTotalInfluence` is a formally sufficient hypothesis for query determinacy and precise PLN/Walley envelopes. Separately, if two DLR completions are exhibited that disagree on a query, then that query has strict width. The Lean development does not prove that failure of the Dobrushin hypothesis entails non-uniqueness or strict width.

7 Worked examples already in the formalization

The codebase now ships both finite toy witnesses and a concrete infinite DLR/PLN contrast crown. The finite two-spin examples remain useful sanity witnesses for strict credal width.

7.1 Two-spin magnet

In `ProjectiveCredal.lean` around lines 1064–1101, the type `TwoSpin`, defined as `Bool × Bool`, carries the two-spin magnet credal set

```
def twoSpinMagnetCredalSet : CredalPrevisionSet TwoSpin :=
  { PrecisePrevision.dirac twoSpinAllDown,
    PrecisePrevision.dirac twoSpinAllUp }
```

The credal set contains two Dirac measures: all spins down, and all spins up. The first-spin-up gamble

```
def twoSpinFirstUpGamble : Gamble TwoSpin :=
  fun s => if s.fst then 1 else 0
```

is rewarded by the all-up Dirac and unrewarded by the all-down Dirac; the disagreement gives the strict width witness. The packaged theorem `twoSpinMagnetEnvelope_nontrivial` (line 1101) closes:

```
theorem twoSpinMagnetEnvelope_nontrivial :
  lowerEnvelope twoSpinMagnetCredalSet twoSpinFirstUpGamble
    < upperEnvelope twoSpinMagnetCredalSet twoSpinFirstUpGamble
```

PLN-side: on this two-Dirac credal, the magnetization query has lower envelope 0, upper envelope 1, and confidence at its floor.

7.2 Zero-temperature aligned magnet

An incremental sibling `twoSpinZeroTemperatureMagnetEnvelope_nontrivial` adds the zero-temperature Gibbs constraint (spins must be aligned), and shows the envelope is still strictly non-trivial: any measure supported on the aligned-spin pair $\{(\downarrow, \downarrow), (\uparrow, \uparrow)\}$ still admits both extremes. The structural constraint by itself does not close down credal width; the Dobrushin condition has to be checked *quantitatively* on the local-influence row-sums.

7.3 Concrete infinite crown: reinforced line versus zero-weight grid

The theorem `reinforcedLineGeometric_strictWidth_and_zeroWeightGrid_collapse` packages the first concrete DLR/PLN contrast. The strict-width side uses `reinforcedLineGeometric_originSpinUp_plnStrictInterval`: a geometrically reinforced half-line has positive PLN strict-width at the origin-spin query. The collapse side uses `gridZeroField_originSpinUp_queryEnvelope_precise_of_dobrushin`: the zero-weight zero-field grid lies in the Dobrushin collapse regime for the analogous origin-spin query. This is an infinite canary, but it is not yet the full two-dimensional low-temperature Ising plus/minus phase-separation theorem.

8 Three kinds of degree of freedom: the synthesis

We now name the three formally-locatable freedom slots, each with the predicate that witnesses it.

(A) Specification-level: localCredal choice. A wider local credal at any cylinder window produces a wider projective limit. The PLN-side question “does my $\langle s, c \rangle$ on query X_q have genuine width?” reduces to “does `hasStrictGlobalWidth spec` on X_q hold?”. The modeller controls this slot by varying `localCredal i`. The slot is finite or infinite depending on whether Ω is.

(B) Completion-level: extreme-point choice. Given a credal set C , the actual world chooses one $P \in C$ (the “true” precise prevision). This freedom is aleatoric. It is formally invisible to $\langle s, c \rangle$ because $\langle s, c \rangle$ is a property of C , not of any specific P . PLN inference rules do not see it; only the verification of a query against actual data sees it.

(C) Phase-transition: DLR non-uniqueness. In infinite Markov-logic specs, the projective limit is the set of DLR completions, which may itself be non-singleton. The slot is opened exactly when `dlrQueryHasStrictWidth` on (M, q) holds for the query q of interest, and closed when `dlrQueryDetermined` on (M, q) holds. The Dobrushin condition `PaperUniformSmallTotalInfluence` is a sufficient condition for the closed case.

Observation 7. *Slot (A) is the freedom the PLN modeller intends to give. Slot (B) is the freedom the world has within the modeller’s spec. Slot (C) is the freedom that the infinite structure of the spec itself produces, without any modeller intent and without any world-side aleatoric residual.*

The three slots are independent. A modeller can pick (A) tight, the world can pick (B) freely, and the spec can still have (C) freedom from a phase transition. The PLN $\langle s, c \rangle$ sees (A) and (C); it does not see (B).

9 What is open

The formal characterization above is honest about which steps are already-proved theorems and which are open. Since the first draft of this section, the generic compact/FIP and Walley projective-natural-extension waist has landed. The remaining work is now narrower:

- **Local finite-window coherence feeding FIP.** The generic compact/FIP waist is present: `ProjectiveLocalLowerPrevisionSpec`. `finiteWindowCompatibleInCarrier` feeds `hasCompatibleCompletion_of_finiteWindowCompatibleInCarrier`. What remains is an honest, non-circular local-coherence predicate that implies finite-window compatibility without assuming a hidden global compatible precise prevision.
- **Polished DLR compact-projective packaging.** The DLR determined/strict-width bridge and the concrete reinforced-line crown exist. The remaining packaging work is to expose the DLR layer as a clean public specialization of the generic projective/Walley interface, with theorem names aligned to the confidence-formula API.
- **Imprecise de Finetti mixing-family package.** The finite-prefix Bernoulli-mixture payoff theorems exist, including `agreement  $\Rightarrow$  width-complement 1` and `disagreement  $\Rightarrow$  width-complement  $< 1$` . What remains is a clean public mixing-family object and, later, full process-law synthesis beyond finite-prefix adapters.
- **True two-dimensional low-temperature Ising separation.** The reinforced-line strict-width crown is a concrete infinite canary. It is not yet the classical 2D low-temperature Ising plus/minus theorem. That theorem would require constructing two distinct infinite-volume DLR completions

satisfying the same local conditional specification and proving that they disagree on an explicit finite-cylinder query.

- **Paper/build metadata refresh.** Line counts and theorem references should be generated from the exact commit or avoided in favor of module and theorem names.

These are not blockers for the dichotomy we have proved; they are the next theorems in the same vein. We name them as open so a reader of this paper is not misled into thinking the infinite half of the characterization is closed.

10 Related work

Walley. The credal-set lower-prevision framework is Walley (1991). Our `LowerPrevision` structure is the Walley structure with constant-preservation, positive homogeneity, and superadditivity, and our `lowerEnvelopePrevision_isCoherent_of_finite` is the classical Walley coherence theorem for finite envelopes.

de Finetti. The de Finetti exchangeability theorem connects exchangeable distributions on infinite sequences to mixtures over IID distributions. The codebase has finite-prefix imprecise de Finetti payoff theorems: mixture agreement forces width-complement 1, and mixture disagreement forces width-complement < 1 on the induced finite-prefix envelope. It also has a projective specialization adapter. What remains open is a polished public mixing-family object and a full process-law synthesis beyond the finite-prefix surface.

Dobrushin–Lanford–Ruelle. The DLR framework characterises infinite Gibbs measures by their finite-volume conditional distributions. The Dobrushin uniqueness condition (uniform smallness of total influence) implies a unique DLR completion. Singla and Domingos have a Markov-logic specific uniformity-of-influence framing. Our `PaperUniformSmallTotalInfluence` is the codified hypothesis.

PLN literature. The two-component truth value $\langle s, c \rangle$ is canonical in the PLN literature (Goertzel et al. *Probabilistic Logic Networks*). The sibling papers §PLN (`papers/xiPLN.tex`) and the PLN Review (`papers/pln-review.tex`) treat the evidence-algebra and formal-rule-coherence axes. The WM-PLN book (`papers/wm-pln-book.tex`) has the informal predecessor of the degrees-of-freedom characterization in its §7.4 (“Degrees of freedom versus forcing”).

Markov logic networks. Richardson and Domingos (Machine Learning, 2006) introduced MLNs as finite log-linear models on first-order logic. The infinite specialisation in `Mettapedia.Logic.MarkovLogicInfinite*` is a recent codex landing (cited inline above).

Imprecise probability and second-order Bayesianity. The PLN tradition of treating c as “confidence-in-strength” has a natural Bayesian second-order reading (a Beta distribution over the true probability), which the file `Mettapedia.Logic.PLNDistributional` formalises. The credal reading we develop here is the Walley-side complement; in the finite case the two readings overlap on the singleton case and diverge on the strict-width case.

11 Conclusion

PLN truth values $\langle s, c \rangle$ are summary statistics of a credal set, and the question of where their information lives reduces, formally, to a dichotomy on the underlying credal set. The codebase landings in `Mettapedia.ProbabilityTheory.ImpreciseProbability.ProjectiveCredal` (the `credalSetDetermines` / `credalSetHasStrictWidth` predicates and their collapse / width theorems) and in `Mettapedia.Logic.MarkovLogicInfiniteCredalBridge` (the DLR-level lift and the Dobrushin hypothesis as a sufficient collapse condition) make this dichotomy a checkable Lean theorem rather than informal commentary.

The three freedom slots — specification, completion, phase-transition — partition the question along orthogonal axes. (A) is the modeller’s freedom. (B) is the world’s freedom. (C) is the structural freedom the spec produces by itself. PLN $\langle s, c \rangle$ sees (A) and (C); it does not see (B). Where it sees (C), the Dobrushin condition is the formally sufficient hypothesis for $\langle s, c \rangle$ to be precise.

The infinite half of the characterization has real open work, named in Sec. 9. We have not pretended otherwise.

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